$P(9 < X < 14) = P(-1 < Z < 0.667)$

$= P(0 < Z < 1) + P(0 < Z < 0.667)$

$= 0.3413 + 0.2486$

$= 0.5899$

The expected battery cells life to have more than 13 hours is 125 x 0.5899 =73.73%

Example 7.30

Weights of fish caught by a traveler are approximately normally distributed with a mean weight of 2.25 kg and a standard deviation of 0.25 kg. What percentage of fish weigh less than 2 kg?

Solution :

We are given mean μ = 2.25 and standard deviation σ = 0.25. Probability that weight of fish is less than 2 kg is $P(X < 2.0)$

When $x = 20$ $Z = \frac{X - \mu}{\sigma} = \frac{2.0 - 2.25}{0.25} = -1.0$

$P(Z < -1.0) = P(Z > 1.0) = 0.5 - 0.3413 = 0.1587$

Therefore 15.87% of fishes weigh less than 2 kg.

Example 7.31

The average daily procurement of milk by village society in 800 litres with a standard deviation of 100 litres. Find out proportion of societies procuring milk between 800 litres to 1000 litres per day.

Solution :

We are given mean μ = 800 and standard deviation σ = 100. Probability that the procurement of milk between 800 litres to 1000 litres per day is

$P(800 < X < 1000) = P\left(\frac{800 - 800}{100} < z < \frac{1000 - 800}{100}\right)$

$= P(0 < Z < 2) = 0.4772$ (table value)

Therefore 47.75 percent of societies procure milk between 800 litres to 1000 litres per day.

Exercise 7.3

1. Define Normal distribution.
2. Define Standard normal variate.
3. Write down the conditions in which the Normal distribution is a limiting case of binomial distribution.
4. Write down any five chief characteristics of Normal probability curve.
5. In a test on 2,000 electric bulbs, it was found that bulbs of a particular make, was normally distributed with an average life of 2,040 hours and standard deviation of 60 hours. Estimate the number of bulbs likely to burn for (i) more than 2,150 hours (ii) less than 1,950 hours (iii) more 1,920 hours but less than 2,100 hours.
6. In a distribution 30% of the items are under 50 and 10% are over 86. Find the mean and standard deviation of the distribution.
7. X is normally distributed with mean 12 and SD 4. Find $P(X \leq 20)$ and $P(0 \leq X \leq 12)$
8. If the heights of 500 students are normally distributed with mean 68.0 inches and standard deviation 3.0 inches , how many students have height (a) greater than 72 inches (b) less than or equal to 64 inches (c) between 65 and 71 inches.
9. In a photographic process, the developing time of prints may be looked upon as a random variable having the normal distribution with a mean of 16.28 seconds and a standard deviation of 0.12 second. Find the probability that it will take less than 16.35 seconds to develop prints.
10. Time taken by a construction company to construct a flyover is a normal variate with mean 400 labour days and standard deviation of 100 labour days. If the company

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